

MA1072/ Matematik 2

İki Katlı İntegraller

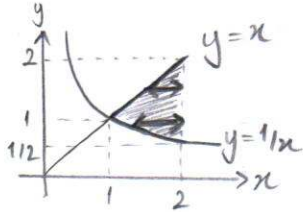
1) $\int_1^2 \int_{1/x}^x \frac{x^2}{y^2} dy dx$

a) integrali verildiği şekilde hesaplayınız.

b) integrasyon sırasını değiştirerek yeni integrali yazınız.

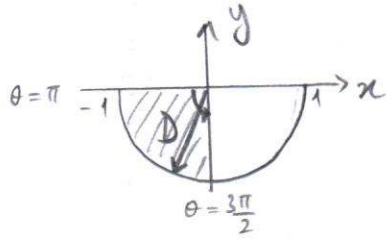
a) $\int_1^2 \int_{1/x}^x \frac{x^2}{y^2} dy dx = \int_1^2 \left[-\frac{x^2}{y} \right]_{1/x}^x dx = \int_1^2 \left[-\frac{x^2}{x} + \frac{x^2}{1/x} \right] dx = -\frac{x^2}{2} + \frac{x^4}{4} \Big|_1^2 = 2 + \frac{1}{4} = \frac{9}{4}$

b) $x=1$ $y=1/x$ $x=2$ $y=x$ $I = \int_{1/2}^1 \int_{1/y}^2 \frac{x^2}{y^2} dx dy + \int_1^2 \int_y^2 \frac{x^2}{y^2} dx dy$



2) $\int_{-1}^0 \int_{-\sqrt{1-x^2}}^0 \frac{2}{1+\sqrt{x^2+y^2}} dy dx$ integralini hesaplayınız.

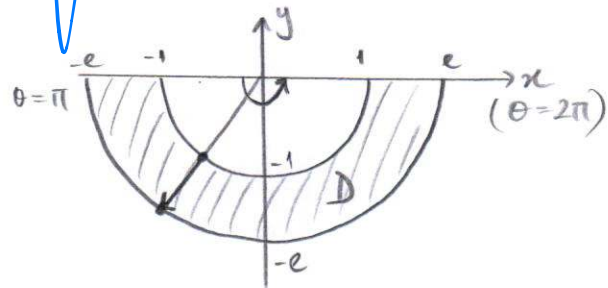
$D = \begin{cases} x=0 \\ x=-1 \end{cases} \begin{cases} y=-\sqrt{1-x^2} \\ y=0 \end{cases}$



$\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases} \Rightarrow dy dx = r dr d\theta$

$I = \int_{\pi}^{3\pi/2} \int_0^1 \frac{2}{1+\sqrt{r^2}} r dr d\theta = 2 \int_{\pi}^{3\pi/2} \int_0^1 \frac{r}{r+1} dr d\theta = 2 \int_{\pi}^{3\pi/2} \int_0^1 \left(1 - \frac{1}{r+1}\right) dr d\theta$
 $= 2 \int_{\pi}^{3\pi/2} (r - \ln(r+1)) \Big|_0^1 d\theta = 2(1 - \ln 2) \left(\frac{3\pi}{2} - \pi\right) = (1 - \ln 2)\pi$

3) D bölgesi, $D = \{(x,y) \in \mathbb{R}^2 \mid 1 \leq x^2 + y^2 \leq e^2, y \leq 0\}$ ile verilsin. Üstten $z = \frac{\ln(x^2 + y^2)}{x^2 + y^2}$ yüzeyi ve alttan xy düzleminde D ile sınırlı cismin hacmini hesaplayınız.



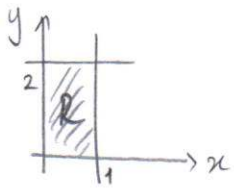
$\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases} \Rightarrow dx dy = r dr d\theta$

$V = \iint_D \frac{\ln(x^2 + y^2)}{x^2 + y^2} dx dy = \int_{\pi}^{2\pi} \int_1^e \frac{\ln(r^2)}{r^2} r dr d\theta$

$= \int_{\pi}^{2\pi} \int_1^e \frac{2\ln r}{r} dr d\theta = \int_{\pi}^{2\pi} \int_1^e 2u du d\theta = \int_{\pi}^{2\pi} \ln^2 r \Big|_1^e d\theta$
 $\left[\begin{matrix} \ln r = u \\ \frac{dr}{r} = du \end{matrix} \right] = \int_{\pi}^{2\pi} d\theta = \theta \Big|_{\pi}^{2\pi} = 2\pi - \pi = \pi$

4) Üstten $z = 10 + x^2 + 3y^2$ paraboloidi ve alttan $z = 0$ da $R: \begin{cases} 0 \leq x \leq 1 \\ 0 \leq y \leq 2 \end{cases}$

dikdörtgeni ile sınırlı bölgenin hacmini hesaplayınız.

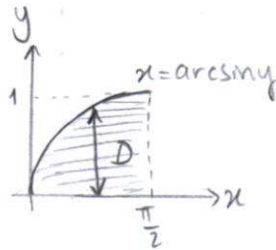


$$V = \iint_R (10 + x^2 + 3y^2) dy dx = \int_0^1 \int_0^2 (10 + x^2 + 3y^2) dy dx$$

$$= \int_0^1 (10y + x^2y + y^3 \Big|_0^2) dx = \int_0^1 (20 + 2x^2 + 8) dx = 28x + \frac{2x^3}{3} \Big|_0^1 = \frac{86}{3}$$

5) $\int_0^1 \int_{\arcsin y}^{\pi/2} e^{\cos x} dx dy$ integralini hesaplayınız.

$$D = \begin{cases} y=0, y=1 \\ x=\arcsin y, x=\frac{\pi}{2} \end{cases}$$

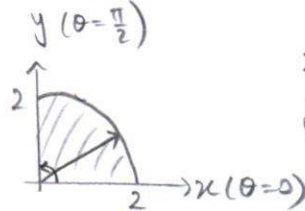


$$\int_0^1 \int_{\arcsin y}^{\pi/2} e^{\cos x} dx dy = \int_0^{\pi/2} \int_0^{\sin x} e^{\cos x} dy dx = \int_0^{\pi/2} e^{\cos x} y \Big|_0^{\sin x} dx = \int_0^{\pi/2} e^{\cos x} \sin x dx$$

$$= -e^{\cos x} \Big|_0^{\pi/2} = e - 1$$

6) $\int_0^2 \int_0^{\sqrt{4-x^2}} \frac{6xy}{(1+2x^2+2y^2)^{3/2}} dy dx$ integralini hesaplayınız.

$$\begin{cases} x=0 & y=\sqrt{4-x^2} \\ x=2 & y=0 \end{cases}$$



$$\begin{cases} x = r \cos \theta \\ y = r \sin \theta \end{cases} \quad \left. \begin{array}{l} \\ \end{array} \right\} dy dx = r dr d\theta$$

$$I = \int_0^{\pi/2} \int_0^2 \frac{3r^3 \sin 2\theta}{(1+2r^2)^{3/2}} dr d\theta = \int_0^{\pi/2} 3 \cdot \frac{1}{3} \sin 2\theta d\theta = -\frac{1}{2} \cos 2\theta \Big|_0^{\pi/2} = -\frac{1}{2} (\cos \pi - \cos 0) = 1$$

$$\int_0^2 \frac{r^3}{(1+2r^2)^{3/2}} dr = \frac{1}{4} \cdot \frac{1}{2} \int_0^2 \frac{u-1}{u^{3/2}} du = \frac{1}{8} \int_0^2 (u^{-1/2} - u^{3/2}) du = \frac{1}{4} (u^{1/2} + u^{-1/2}) \Big|_0^2 = \frac{1}{4} ((1+2r^2)^{1/2} + (1+2r^2)^{-1/2}) \Big|_0^2$$

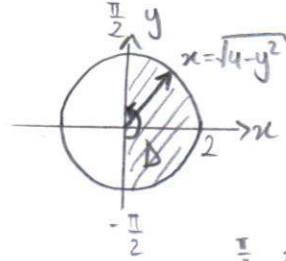
$$= \frac{1}{3}$$

$$\left[\begin{array}{l} u = 1+2r^2 \Rightarrow r^2 = \frac{u-1}{2} \\ du = 4r dr \end{array} \right]$$

7) D bölgesi, $x = \sqrt{4-y^2}$ ve y-ekseni ile sınırlı bir bölge olmak üzere

$\iint_D e^{-x^2-y^2} dA$ integralini hesaplayınız

$$D = \left. \begin{aligned} x &= \sqrt{4-y^2} \Rightarrow x^2+y^2=4 \\ x &= 0 \end{aligned} \right\}$$

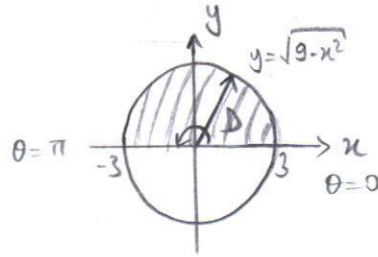


$$\left. \begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \right\} \begin{aligned} x^2+y^2 &= r^2 \\ dA &= r dr d\theta \end{aligned}$$

$$\begin{aligned} \iint_D e^{-x^2-y^2} dA &= \int_{-\pi/2}^{\pi/2} \int_0^2 e^{-r^2} r dr d\theta \quad \begin{aligned} r^2 &= u \\ 2r dr &= du \end{aligned} \Rightarrow \int_{-\pi/2}^{\pi/2} \int_0^2 \frac{e^{-u}}{2} du d\theta = \int_{-\pi/2}^{\pi/2} -\frac{1}{2} e^{-r^2} \Big|_0^2 d\theta \\ &= \int_{-\pi/2}^{\pi/2} \left(-\frac{1}{2} e^{-4} + \frac{1}{2} \right) d\theta = -\frac{1}{2} e^{-4} \theta + \frac{\theta}{2} \Big|_{-\pi/2}^{\pi/2} \\ &= \frac{\pi}{2} (1 - e^{-4}) \end{aligned}$$

8) $\int_{-3}^3 \int_0^{\sqrt{9-x^2}} \sin(x^2+y^2) dy dx$ integralini hesaplayınız.

$$D = \left. \begin{aligned} y &= \sqrt{9-x^2} \Rightarrow x^2+y^2=9 \\ y &= 0 \\ x &= 3, x = -3 \end{aligned} \right\}$$

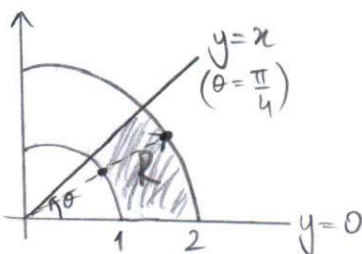


$$\left. \begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \right\} \begin{aligned} x^2+y^2 &= r^2 \\ dy dx &= r dr d\theta \end{aligned}$$

$$\begin{aligned} \int_{-3}^3 \int_0^{\sqrt{9-x^2}} \sin(x^2+y^2) dy dx &= \int_0^{\pi} \int_0^3 \sin(r^2) r dr d\theta \quad \begin{aligned} u &= r^2 \\ du &= 2r dr \end{aligned} \Rightarrow \int_0^{\pi} \int_0^3 \frac{\sin u}{2} du d\theta \\ &= \int_0^{\pi} -\frac{1}{2} \cos r^2 \Big|_0^3 d\theta = \int_0^{\pi} \left(-\frac{1}{2} \cos 9 + \frac{1}{2} \right) d\theta = \left(-\frac{1}{2} \cos 9 \right) \theta + \frac{\theta}{2} \Big|_0^{\pi} \\ &= \frac{\pi}{2} (1 - \cos 9) \end{aligned}$$

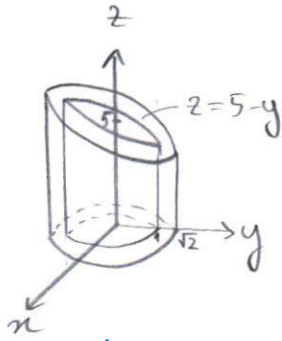
9) R bölgesi $\{(x,y) \in \mathbb{R}^2 \mid 1 \leq x^2+y^2 \leq 4, 0 \leq y \leq x\}$ olmak üzere,

$\iint_R \arctan\left(\frac{y}{x}\right) dA$ integralini hesaplayınız.

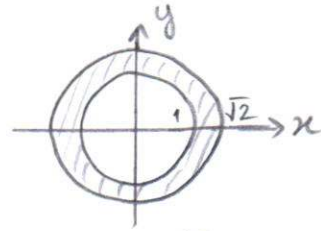


$$\begin{aligned} \iint_R \arctan\left(\frac{y}{x}\right) dA &= \int_0^{\pi/4} \int_1^2 \underbrace{\arctan(\tan \theta)}_{=\theta} r dr d\theta \\ &= \int_0^{\pi/4} \theta \frac{r^2}{2} \Big|_1^2 d\theta = \int_0^{\pi/4} \frac{3\theta}{2} d\theta = \frac{3\theta^2}{4} \Big|_0^{\pi/4} = \frac{3\pi^2}{64} \end{aligned}$$

- 10) $z+y=5$ düzleminin altında, xy -düzleminin üstünde $x^2+y^2=2$ ile $x^2+y^2=1$ silindirleri arasında kalan cismin hacmini bulunuz.



İz düşüm bölgesi:



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$dx dy = r dr d\theta$$

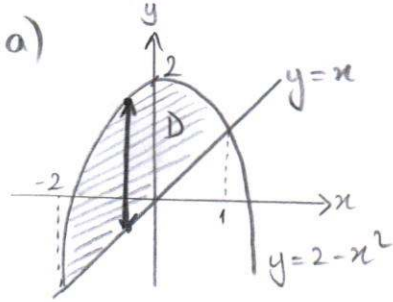
$$V = \int_0^{2\pi} \int_1^{\sqrt{2}} (5 - r \sin \theta) r dr d\theta = \int_0^{2\pi} \left(\frac{5}{2} + \frac{1-2\sqrt{2}}{3} \sin \theta \right) d\theta$$

$$= \frac{5}{2} \theta - \frac{1-2\sqrt{2}}{3} \cos \theta \Big|_0^{2\pi} = 5\pi$$

- 11) D bölgesi $y=2-x^2$ parabolü ve $y=x$ doğrusu ile sınırlı bölge olsun.

a) Üstten $z=x^2$ yüzeyi ve alttan xy -düzleminde D bölgesi tarafından sınırlanan cismin hacmini bulunuz.

b) $f(x,y)=x^2$ fonksiyonunun D bölgesi üzerinde ortalama değerini bulunuz.



$$2 - x^2 = x \Rightarrow x^2 + x - 2 = 0 \Rightarrow x = -2, x = 1$$

$$V = \iint_D x^2 dx dy = \int_{-2}^1 \int_x^{2-x^2} x^2 dy dx = \int_{-2}^1 x^2 y \Big|_x^{2-x^2} dx$$

$$= \int_{-2}^1 x^2 (2 - x^2 - x) dx = \int_{-2}^1 (-x^4 - x^3 + 2x^2) dx$$

$$= -\frac{x^5}{5} - \frac{x^4}{4} + \frac{2}{3} x^3 \Big|_{-2}^1 = \frac{63}{20}$$

b) D üzerinde f nin ortalama değeri = $\frac{1}{D \text{ nin alanı}} \iint_D f(x,y) dA$

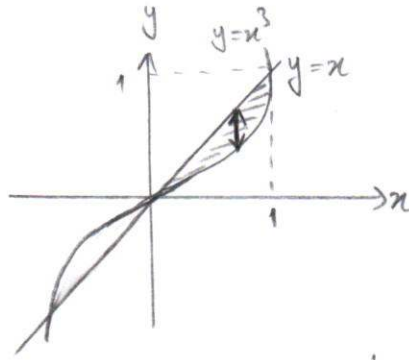
$$D \text{ bölgesinin alanı} = A = \iint_D dy dx = \int_{-2}^1 \int_x^{2-x^2} dy dx = \int_{-2}^1 (2 - x^2 - x) dx$$

$$= 2x - \frac{x^3}{3} - \frac{x^2}{2} \Big|_{-2}^1 = \frac{9}{2}$$

$$\text{Ort. değer} = \frac{V}{A} = \frac{63}{20} \cdot \frac{2}{9} = \frac{7}{10}$$

12) $\int_0^1 \int_{x^3}^x \frac{1}{1+x^4} dx dy$ integralini hesaplayınız.

$\left. \begin{array}{l} x=y \\ y=0 \\ x=\sqrt[3]{y} \\ y=1 \end{array} \right\}$



$$I = \int_0^1 \int_{x^3}^x \frac{1}{1+x^4} dy dx = \int_0^1 \frac{x-x^3}{1+x^4} dx$$

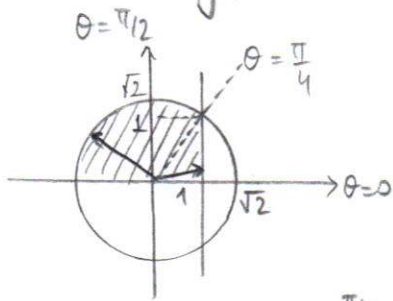
$$= \int_0^1 \frac{x(1-x^2)}{1+x^4} dx$$

$x^2 = t$
 $2x dx = dt$
 $x \rightarrow 0 \quad t \rightarrow 0$
 $x \rightarrow 1 \quad t \rightarrow 1$

$$I = \frac{1}{2} \int_0^1 \frac{1-t}{1+t^2} dt = \frac{1}{2} \int_0^1 \frac{1}{1+t^2} dt - \frac{1}{2} \int_0^1 \frac{t}{1+t^2} dt = \frac{1}{2} \arctan t \Big|_0^1 - \frac{1}{4} \ln(1+t^2) \Big|_0^1$$

$$= \frac{1}{2} \left(\frac{\pi}{4} - 0 \right) - \frac{1}{4} \ln 2 = \frac{\pi - \ln 4}{8}$$

13) $D = \begin{cases} x^2 + y^2 \leq 2 \\ x \leq 1 \\ y \geq 0 \end{cases}$

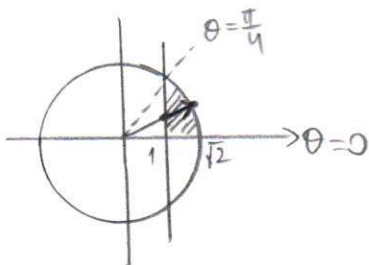


bölgesinde $I = \iint_D x dA$ integralinin sınırlarını kutupsal dönüşüm ile yazınız.

$$\left. \begin{array}{l} x = r \cos \theta \\ y = r \sin \theta \\ dx dy = r dr d\theta \end{array} \right\} \begin{array}{l} x=1 \Rightarrow r \cos \theta = 1 \Rightarrow r = \frac{1}{\cos \theta} = \sec \theta \\ x^2 + y^2 = 2 \Rightarrow r = \sqrt{2} \end{array}$$

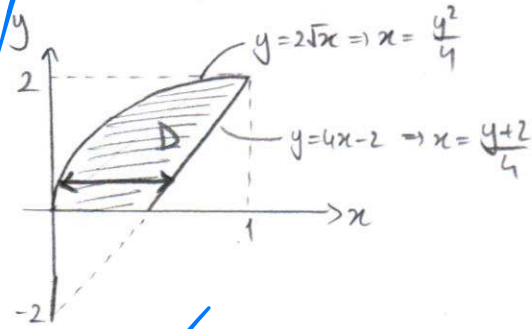
$$I = \iint_D x dA = \int_0^{\pi/4} \int_0^{\sec \theta} r \cos \theta r dr d\theta + \int_{\pi/4}^{\pi/2} \int_0^{\sqrt{2}} r \cos \theta r dr d\theta$$

$D = \begin{cases} x^2 + y^2 \leq 2 \\ x \geq 1 \\ y \geq 0 \end{cases}$ bölgesinde aynı integral;



$$I = \iint_D x dA = \int_0^{\pi/4} \int_{\sec \theta}^{\sqrt{2}} r \cos \theta r dr d\theta$$

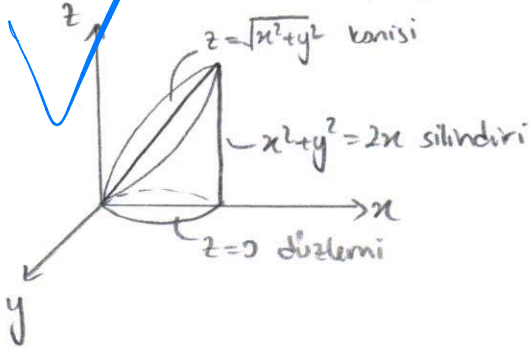
- 14) $z=16-x^2-y^2$ yüzeyinin altında ve $z=0$ 'da, $y=2\sqrt{x}$, $y=4x-2$ ve x -eksenini ile sınırlı D bölgesinin üstünde bulunan cismin hacmini veren integrali yazınız.



$$V = \iint_D (16-x^2-y^2) dx dy$$

$$= \int_0^2 \int_{\frac{y+2}{4}}^{\frac{y^2}{4}} (16-x^2-y^2) dx dy$$

- 15) $x^2+y^2=2x$, $z=0$, $z=\sqrt{x^2+y^2}$ yüzeyleri ile sınırlandırılmış cismin hacmini veren integrali yazınız.



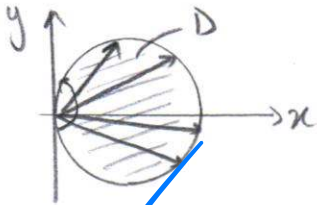
$$x^2+y^2-2x+1-1=0 \Rightarrow (x-1)^2+y^2=1$$

$$V = \iint_D f(x,y) dx dy = \iint_D \sqrt{x^2+y^2} dx dy$$

yükseklik koniden, taban alanı silindirden

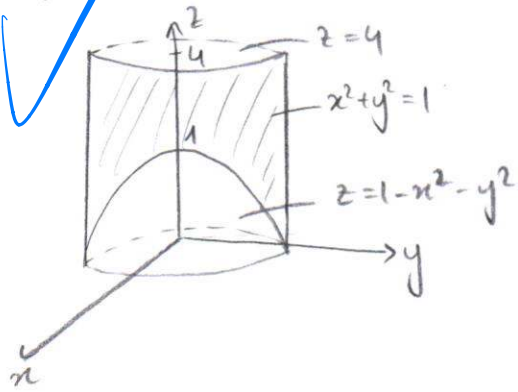
$$\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases} \begin{cases} dx dy = r dr d\theta \\ \sqrt{x^2+y^2} = r \end{cases}$$

$$\begin{cases} x^2+y^2=2x \\ r^2=2r\cos\theta \end{cases} \Rightarrow r=2\cos\theta$$

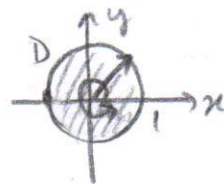


$$V = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r dr d\theta$$

- 16) $x^2+y^2=1$ silindirin içinde, $z=4$ düzleminin altında, $z=1-x^2-y^2$ parabolünün üzerinde kalan cismin hacmini bulunuz.



$$D: x^2+y^2=1$$



$$\begin{cases} x=r\cos\theta \\ y=r\sin\theta \end{cases} \begin{cases} dx dy = r dr d\theta \\ x^2+y^2=r^2 \end{cases}$$

$$V = \iint_D (4 - (1-x^2-y^2)) dx dy = \int_0^{2\pi} \int_0^1 (3+r^2) r dr d\theta = \int_0^{2\pi} \left(\frac{3}{2} r^2 + \frac{r^4}{4} \right) \Big|_0^1 d\theta = \int_0^{2\pi} \frac{7}{4} d\theta = \frac{7\pi}{2}$$

17)

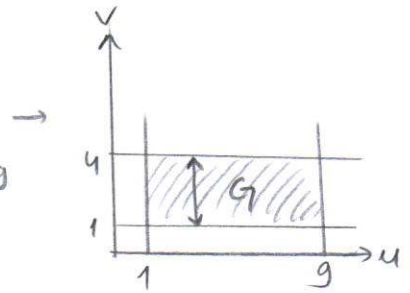
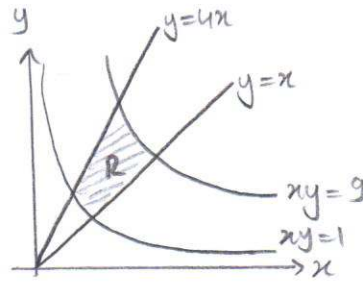
$$R = \begin{cases} xy=1 \\ xy=9 \\ y=x \\ y=4x \end{cases}$$

epirlerinin 1. bölgede sınırladığı bölge ise

$$\iint_R \left(\sqrt{\frac{y}{x}} + \sqrt{xy} \right) dx dy = ?$$

$$\begin{cases} xy=u \\ \frac{y}{x}=v \end{cases}$$

$$\begin{array}{ll} R & G \\ xy=1 & \rightarrow u=1 \\ xy=9 & \rightarrow u=9 \\ y=x & \rightarrow v=1 \\ y=4x & \rightarrow v=4 \end{array}$$



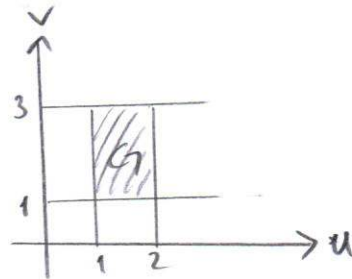
$$J(u,v) = \frac{1}{\begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix}} = \frac{1}{\begin{vmatrix} y & x \\ -\frac{y}{x^2} & \frac{1}{x} \end{vmatrix}} = \frac{1}{\frac{y}{x} + \frac{y}{x}} = \frac{1}{\frac{2y}{x}} = \frac{1}{2v}$$

$$\begin{aligned} \iint_R \left(\sqrt{\frac{y}{x}} + \sqrt{xy} \right) dx dy &= \int_1^9 \int_1^4 \left(\sqrt{v} + \sqrt{u} \right) \frac{1}{2v} dv du = \int_1^9 \left(\sqrt{v} + \frac{1}{2} \sqrt{u} \ln v \right) \Big|_1^4 du \\ &= \int_1^9 (1 + \ln 2 \sqrt{u}) du = u + \ln 2 \cdot \frac{2}{3} u^{3/2} \Big|_1^9 = 8 + \frac{52}{3} \ln 2 \end{aligned}$$

18) $y=x^2$, $y=2x^2$, $x=y^2$, $x=3y^2$ parabolleri ile sınırlanmış bölgenin alanını iki katlı integral yardımıyla hesaplayınız.

$$\begin{cases} \frac{y}{x^2}=u \\ \frac{x}{y^2}=v \end{cases}$$

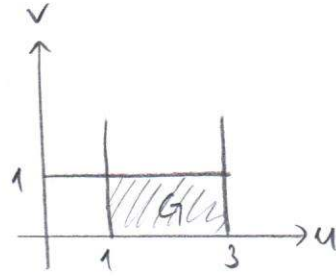
$$\begin{array}{ll} D & G \\ \frac{y}{x^2}=1 & \rightarrow u=1 \\ \frac{y}{x^2}=2 & \rightarrow u=2 \\ \frac{x}{y^2}=1 & \rightarrow v=1 \\ \frac{x}{y^2}=3 & \rightarrow v=3 \end{array}$$



$$\begin{aligned} J(u,v) &= \frac{1}{\begin{vmatrix} -\frac{2y}{x^3} & \frac{1}{x^2} \\ \frac{1}{y^2} & -\frac{2x}{y^3} \end{vmatrix}} = \frac{x^2 y^2}{3} = \frac{1}{3u^2 v^2} \Rightarrow A = \iint_G \frac{1}{3u^2 v^2} du dv = \frac{1}{3} \int_1^2 \int_1^3 \frac{1}{u^2 v^2} dv du \\ &= \frac{1}{3} \cdot \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{9} \end{aligned}$$

19) Eger D , $x+y=1$, $x+y=3$, $x-y=0$, $x-y=1$ doğruları ile oluşturulmuş bölge ise $\iint_D \frac{x^2-y^2}{\sqrt{1+3(x-y)^2}} dA$ integralini hesaplayınız.

$$\begin{aligned} \left. \begin{aligned} u &= x+y \\ v &= x-y \end{aligned} \right\} & \begin{array}{c|c} D & G \\ \hline x+y=1 & \rightarrow u=1 \\ x+y=3 & \rightarrow u=3 \\ x-y=0 & \rightarrow v=0 \\ x-y=1 & \rightarrow v=1 \end{array} \end{aligned}$$



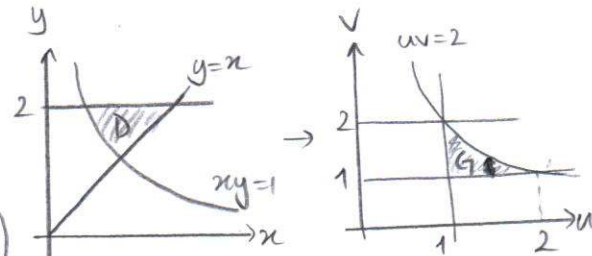
$$J(u,v) = \frac{1}{\begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix}} = -\frac{1}{2}$$

$$\begin{aligned} \iint_D \frac{x^2-y^2}{\sqrt{1+3(x-y)^2}} dA &= \iint_G \frac{uv}{\sqrt{1+3v^2}} \frac{1}{2} du dv = \frac{1}{2} \int_0^1 \int_1^3 \frac{uv}{\sqrt{1+3v^2}} du dv = \frac{1}{2} \int_0^1 4 \cdot \frac{v}{\sqrt{1+3v^2}} dv \\ &= 2 \cdot \frac{1}{3} \sqrt{1+3v^2} \Big|_0^1 = \frac{2}{3} \end{aligned}$$

20) $\iint_D \sqrt{\frac{y}{x}} e^{\sqrt{xy}} dx dy = ?$

$$\left. \begin{aligned} u &= \sqrt{xy} \\ v &= \sqrt{\frac{y}{x}} \end{aligned} \right\} \begin{array}{c|c} D & G \\ \hline x = \frac{1}{y} \Rightarrow xy=1 & \rightarrow u=1 \\ x=y & \rightarrow v=1 \\ y=2 & \rightarrow uv=2 \\ y=1 \text{ (nokta)} & \end{array}$$

$v = \sqrt{\frac{y}{x}} \Rightarrow \frac{y}{x} = v^2 \Rightarrow y = xv^2$



$$u^2 = xy, \quad v^2 = \frac{y}{x} \Rightarrow u^2 v^2 = y^2, \quad \frac{u^2}{v^2} = x^2 \Rightarrow y = uv, \quad x = \frac{u}{v}$$

$$J(u,v) = \begin{vmatrix} \frac{1}{v} & -\frac{u}{v^2} \\ v & u \end{vmatrix} = \frac{2u}{v}$$

$$\begin{aligned} \iint_D \sqrt{\frac{y}{x}} e^{\sqrt{xy}} dx dy &= \iint_G v e^u \frac{2u}{v} du dv = \int_1^2 \int_1^{2/v} 2u e^u dv du = 2 \int_1^2 v u e^u \Big|_1^{2/v} du \\ &= 2 \int_1^2 (2e^u - u e^u) du = 2 (2e^u - u e^u) \Big|_1^2 = 2(e-2) \end{aligned}$$